



FINAL REPORT

**The Art of Scientific Computing:
Respiratory Rates Determining breathing rates through
electrocardiogram data analysis ECG-Derived
Respiratory Rate Estimation**

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Summary

This report investigates whether respiratory-related information can be estimated from long-duration electrocardiogram (ECG) recordings. The project uses the PhysioNet Apnea-ECG database and focuses on developing a clear and reproducible Python workflow for reading ECG data, interpreting QRS and apnea annotations, extracting R–R intervals, and analysing their variation over time.

At the current stage, the ECG waveform and annotation files have been successfully loaded and aligned on a common time axis. Short ECG segments have been visualised together with the supplied QRS annotations to confirm that the annotation locations correspond closely to the R peaks. The next stage of the project is to construct the R–R interval time series and investigate whether respiratory-related periodic behaviour can be identified and used to estimate respiratory rate. Further analysis will compare normal and apnea-labelled periods and examine the reliability and limitations of the method.

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Chapter 1

Introduction

1.1 Background

Respiratory rate is the number of complete breathing cycles per minute and is an important physiological measurement for monitoring respiratory condition [1]. Changes in breathing rate may provide useful information about a person’s respiratory status, especially during long-term monitoring.

Respiration is normally measured using dedicated respiratory sensors. However, electrocardiogram (ECG) signals are also commonly recorded over long periods and contain detailed information about cardiac activity. This raises an interesting question: whether useful respiratory-related information can be obtained indirectly from ECG recordings.

This project uses the PhysioNet Apnea-ECG database, which contains long-duration ECG recordings together with QRS and apnea annotations [2]. The availability of both the ECG waveform and its annotations makes the dataset suitable for developing and testing a reproducible signal analysis workflow.

From a scientific computing perspective, this problem involves several stages. The raw ECG data must first be read and interpreted correctly, the annotations must be aligned with the waveform, useful features must then be extracted, and the resulting time-series data must be analysed to identify possible respiratory-related patterns.

1.2 ECG and Respiratory Information

An electrocardiogram records the electrical activity of the heart over time [4]. A typical ECG waveform contains several characteristic components, including the P wave, QRS complex and T wave. In this project, the QRS complex is especially important because the R peak provides a useful time marker for each heartbeat.

If two successive R peaks occur at times t_i and t_{i+1} , the corresponding R–R interval is defined as

$$RR_i = t_{i+1} - t_i. \quad (1.1)$$

The R–R interval sequence describes how the time between successive heartbeats changes over the recording. Since later analysis depends directly on these intervals, reliable QRS identification is important. Errors in QRS detection can affect heart-rate variability measurements and other quantities derived from R–R intervals [5].

The main idea of this project is to investigate whether variation in the R–R interval sequence contains useful respiratory-related information. The analysis therefore follows the general pathway

ECG waveform $\rightarrow$ QRS / R peaks $\rightarrow$ R–R intervals $\rightarrow$ time-series analysis $\rightarrow$ respiratory-related information.

This task is not straightforward because R–R intervals may also vary because of other physiological changes, abnormal beats, noise, and differences in ECG morphology. Therefore, changes in R–R intervals cannot simply be interpreted as respiration, and further time-series analysis is required.

1.3 Project Aim

The main aim of this project is to develop a clear and reproducible Python workflow for extracting respiratory-related information from ECG recordings.

The first stage of the project focuses on correctly reading and visualising the ECG waveform together with the supplied QRS and apnea annotations. The QRS annotations are then used to obtain the timing of successive heartbeats and to construct an R–R interval time series.

The next stage is to analyse the variation of the R–R intervals and investigate whether useful respiratory-related patterns can be identified. In particular, the project aims to determine whether respiratory rate can be estimated from information derived from the ECG.

After the main ECG-to-respiratory analysis has been established, the extracted features will also be compared between normal and apnea-labelled periods. Additional time-series analysis or classification methods may be considered as extensions if they provide useful information beyond the main analysis.

Chapter 2

ECG Background and Data Description

2.1 ECG Signal and QRS Complex

An electrocardiogram (ECG) records the electrical activity of the heart over time. A typical ECG waveform contains several characteristic components, including the P wave, QRS complex and T wave [4]. The P wave is associated with atrial depolarisation, the QRS complex with ventricular depolarisation, and the T wave with ventricular repolarisation.

Figure 2.1 illustrates the main components of a typical ECG waveform and the definition of the R–R interval.

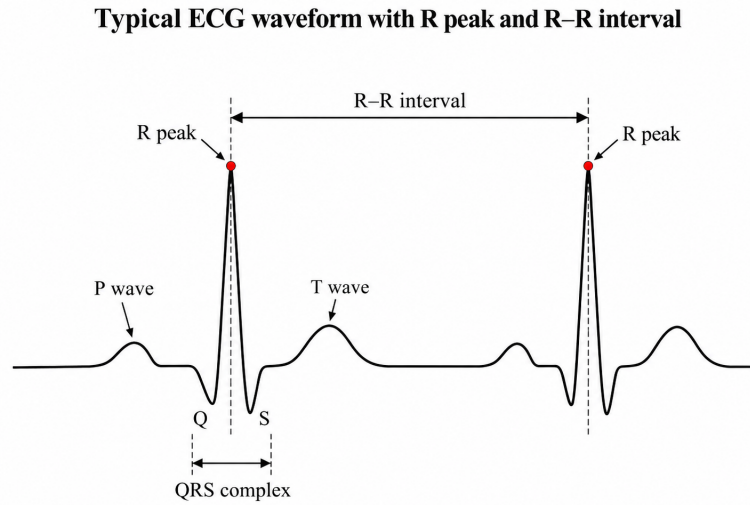


Figure 2.1: Schematic ECG waveform showing the P wave, QRS complex, T wave, R peaks and R–R interval. The R peaks are used as heartbeat timing references in this project.

For this project, the QRS complex is the most important part of the ECG waveform because the R peak provides a clear time marker for each heartbeat. If two successive R peaks occur at times t_i and t_{i+1} , the corresponding R–R interval is defined as

$$RR_i = t_{i+1} - t_i. \quad (2.1)$$

The R–R interval sequence describes how the timing between successive heartbeats changes over time. Since the later respiratory analysis depends on this sequence, reliable identification of QRS complexes is important. Errors in QRS detection can affect quantities derived from the R–R intervals, including heart-rate variability measures [5].

Breathing can also influence the timing between heartbeats. Therefore, changes in the R–R interval sequence may contain useful respiratory-related information. In this project, the R–R intervals are used as the main ECG-derived time series for later respiratory analysis.

2.2 Apnea-ECG Database

The data used in this project are obtained from the PhysioNet Apnea-ECG Database [2]. The database contains long-duration ECG recordings together with metadata and annotation files. These annotations make it possible to study both heartbeat timing and sleep-apnea-labelled periods within the same recording.

The analysis begins with record **a01**. Its header file contains the following information:

```
a01 1 100 2957000
a01.dat 16 200 12 0 -12 5827 0 ECG
```

The first line indicates that record **a01** contains one signal sampled at 100 Hz, with a total of 2,957,000 samples. Therefore, the total recording length is approximately

$$\frac{2,957,000}{100} = 29,570 \text{ s}, \quad (2.2)$$

which is approximately 8.21 hours.

Because the recording is several hours long, plotting the complete ECG waveform at once would make individual beats impossible to inspect clearly. Short time windows are therefore used for waveform inspection, while longer sections of the signal are used later for time-series analysis.

2.3 WFDB File Structure and Annotations

The Apnea-ECG data are stored using the WFDB format. Several related files are used for each record [3]:

- **.dat**: contains the ECG waveform data;
- **.hea**: contains the header and metadata required to interpret the signal;
- **.qrs**: contains annotations for detected QRS complexes;
- **.apn**: contains annotations for sleep-apnea periods.

The waveform and annotation files need to be interpreted on a common time axis. The QRS annotation locations are stored as sample indices and can be converted to time in seconds using

$$t = \frac{n}{f_s}, \quad (2.3)$$

where n is the sample index and f_s is the sampling frequency.

For record **a01**, the apnea annotation positions occur every 6000 samples. Since the sampling frequency is 100 Hz,

$$\frac{6000}{100} = 60 \text{ s}. \quad (2.4)$$

Therefore, one apnea annotation is provided for each minute of the recording. The annotation symbols used in record **a01** include **N** for normal-labelled periods and **A** for apnea-labelled periods.

The three main sources of timing information therefore operate on different time scales. The ECG waveform is sampled at 100 Hz, QRS annotations occur approximately once per heartbeat, and apnea annotations occur once per minute. Careful conversion between samples, seconds and minutes is therefore required before these data can be compared.

2.4 Initial ECG and Annotation Inspection

Before carrying out respiratory analysis, representative ECG segments are first visualised to check that the waveform and annotation files have been read correctly.

Figure 2.2 shows the first 10 seconds of record **a01**. A short time interval is used so that the individual ECG cycles remain visible. In this segment, repeated large peaks can be clearly observed, corresponding to the R peaks of successive heartbeats.

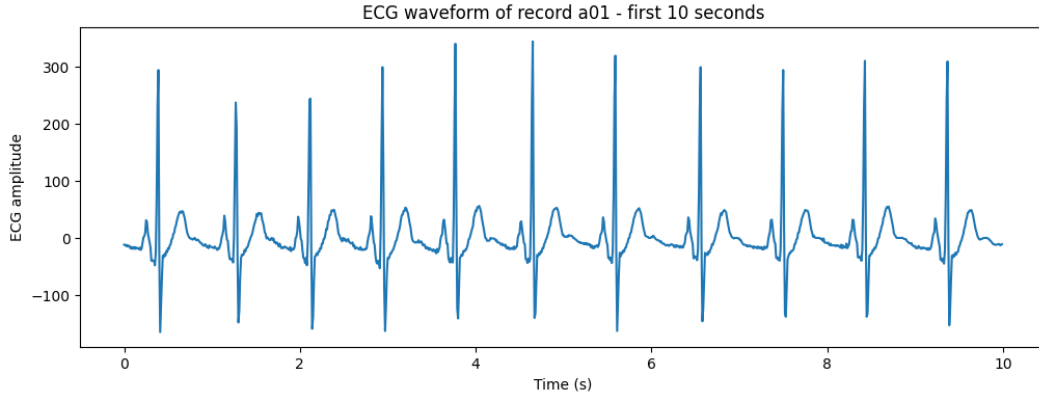


Figure 2.2: ECG waveform of record **a01** over the first 10 seconds. The short window allows individual heartbeat cycles and R peaks to be inspected clearly.

The supplied QRS annotations are then plotted on the same ECG segment, as shown in Figure 2.3. The annotation locations align closely with the large R peaks in the waveform. This provides a basic validation that the ECG waveform and QRS annotation files have been read consistently.

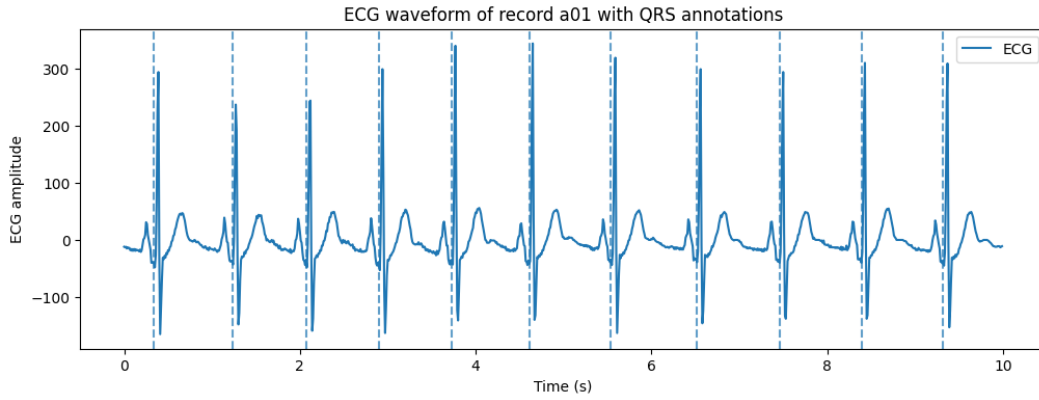


Figure 2.3: ECG waveform of record **a01** with supplied QRS annotations. The QRS annotation locations align closely with the R peaks of the ECG waveform.

The apnea annotations are also converted onto the ECG time axis. However, since each apnea label represents a full 60-second interval, plotting several minutes of ECG together with all QRS annotations produces a very dense figure. Shorter windows around selected apnea-labelled periods are therefore used later when the normal and apnea periods are compared.

Chapter 3

Methodology

This chapter describes the processing workflow used to move from the raw ECG recording to ECG-derived time-series information. The current stage of the analysis focuses on loading and checking the ECG data, reading the supplied QRS annotations, and constructing the R–R interval sequence. Later stages will use this sequence to investigate respiratory-related information.

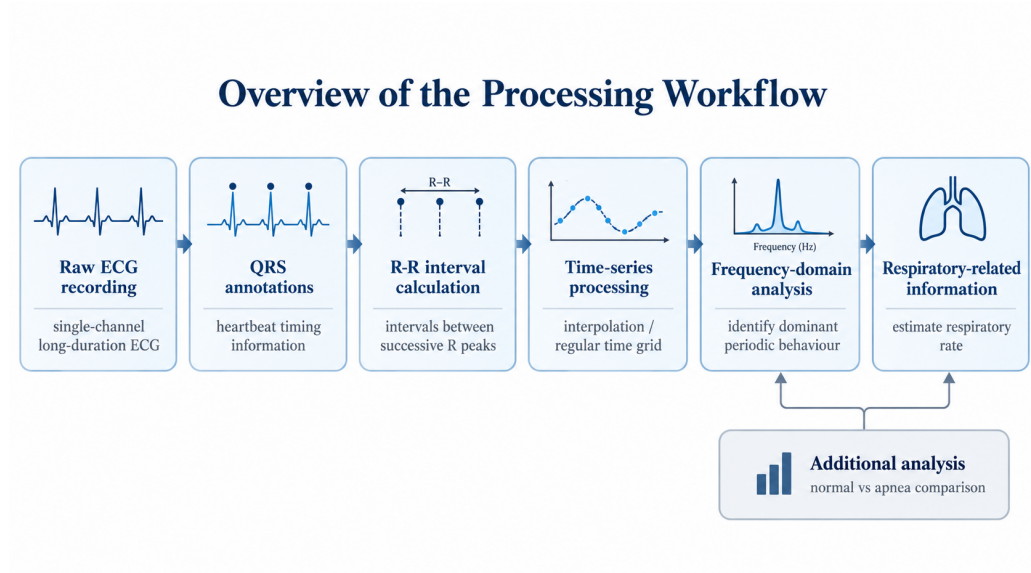


Figure 3.1: Overview of the processing workflow used in this project, from the raw ECG recording to R–R interval extraction and respiratory-related analysis.

3.1 ECG Reading and Visualisation

The ECG waveform is read from the Apnea-ECG record using Python. The sampling frequency is obtained from the corresponding header file. For record `a01`, the sampling frequency is

$$f_s = 100 \text{ Hz.}$$

This means that 100 ECG samples are recorded every second.

A time axis is constructed from the sample indices using

$$t_n = \frac{n}{f_s}, \quad (3.1)$$

where n is the sample index and f_s is the sampling frequency.

Because the full recording is more than eight hours long, plotting the entire signal at once would make individual ECG cycles difficult to distinguish. Short time windows are therefore used for visual inspection of the waveform.

At the current stage, the raw ECG values are used directly for visualisation. No conversion to physical units is applied unless the scaling information from the WFDB header is explicitly used.

3.2 QRS and R-Peak Information

The supplied `.qrs` annotation file is read using the WFDB tools. Each annotation gives the sample position of a detected QRS complex.

The annotation sample positions are converted to time in seconds using

$$t_i = \frac{n_i}{f_s}, \quad (3.2)$$

where n_i is the sample index of the i th QRS annotation.

Before the annotations are used for further analysis, their positions are visually compared with the ECG waveform. In the inspected segments, the annotation positions align closely with the large R peaks. This provides a basic check that the QRS annotations and ECG waveform are being interpreted on the same time axis.

The supplied QRS annotations are therefore used as the initial reference for heartbeat timing in the following R–R interval calculation.

3.3 R–R Interval Calculation

For successive R peaks occurring at times t_i and t_{i+1} , the R–R interval is calculated as

$$RR_i = t_{i+1} - t_i. \quad (3.3)$$

The supplied QRS annotations are first converted from sample indices to time in seconds. The difference between consecutive QRS annotation times is then used to obtain the R–R interval sequence.

To associate each R–R interval with a time position, the midpoint between the two corresponding R peaks is used:

$$t_{RR,i} = \frac{t_i + t_{i+1}}{2}. \quad (3.4)$$

This produces a time series of R–R intervals that describes how the time between successive heartbeats changes throughout the recording.

Because the full record contains a large number of heartbeats, short time windows are also inspected separately. A five-minute segment is used for initial inspection because it is long enough to show changes in the R–R interval sequence while remaining easy to visualise.

3.4 R–R Interval Time-Series Processing

The R–R interval sequence is not sampled at regular time intervals because successive heartbeats do not occur at exactly equal time spacing. This is an important issue for later frequency-domain analysis.

Before respiratory-related frequency analysis is applied, the R–R interval series will therefore be represented on a regular time grid using interpolation. The interpolated signal can then be analysed using methods that assume evenly spaced samples.

3.5 Respiratory-Related Analysis

The main aim of the later analysis is to investigate whether periodic variation in the R–R interval sequence contains respiratory-related information.

A frequency-domain method will be considered to identify dominant periodic behaviour in the processed R–R signal. If a dominant respiratory-related frequency f_r is identified in Hz, the corresponding respiratory rate can be calculated as

$$RR_{\text{resp}} = 60f_r, \quad (3.5)$$

where RR_{resp} is measured in breaths per minute.

The choice of preprocessing method, analysis window and frequency range will be tested before the final respiratory-rate estimation method is selected.

3.6 Additional Analysis

Additional analyses will only be carried out after the main ECG-to-respiratory pipeline has been established and tested.

3.6.1 Normal and Apnea-Labelled Periods

The respiratory-related behaviour of the R–R sequence may later be compared between normal-labelled and apnea-labelled intervals.

3.6.2 Apnea Prediction

Apnea prediction will only be considered if the main respiratory-rate analysis is working reliably.

3.6.3 Alternative Time-Series Methods

Alternative time-series or time-frequency methods may be tested if they provide additional information beyond the baseline analysis.

3.6.4 Regression or Classification

Regression or classification methods will be treated as optional extensions rather than part of the main analysis.

Chapter 4

Discussion

4.1 Main Findings

4.2 Reliability of the Analysis

4.3 Interpretation of Respiratory Information

4.4 Additional Analyses

4.5 Limitations

Important limitations may include:

- QRS annotations may contain errors;
- ECG morphology varies between records;
- R–R interval changes are influenced by factors other than respiration;
- parameter choices such as window length may influence the result;
- direct respiratory ground truth may not be available for every record.

Chapter 5

Conclusion

Appendix A

Program Workflow

Appendix B

Additional Figures and Tables

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